

UFOR 271B

ENVIRONMENTAL SCIENCE

From the Bluff to the Bayou

Environmental Science for Resilient Communities

Southern University and A&M College • Baton Rouge, Louisiana

Central Course Question: How can environmental science help South Louisiana communities become healthier, more equitable, and more resilient?

Course item	Information
Credit and contact hours	3 credit hours • Lecture: 2 hours • Laboratory/field: 2 hours
Prerequisite	SBIO 101B
Term and format	[Term and year] • [In person / hybrid / asynchronous]
Meeting information	[Days, times, classroom, and laboratory/field location]
Instructor	[Name, office, email, telephone, office hours]

WORKING SYLLABUS • Final institutional language, dates, and instructor information must be added before publication.

Course Description

An introductory course to basic principles of science and ecology, environmental protection and sustainability, natural and human systems, population, resource issues, pollution, environmental ethics, policy, economics, and society. The course addresses the sustainable development imperative and includes required field-trip and laboratory exercises.

In this section, students investigate environmental systems through South Louisiana cases while developing transferable skills in field observation, data literacy, geospatial reasoning, environmental sampling, collaborative research, risk communication, and evidence-based decision-making. The course connects scientific foundations to environmental justice, community health, urban forestry, coastal resilience, and workforce preparation.

Course Design Commitments

- Environmental justice and community well-being are coursewide lenses, not isolated topics.
- Southern University's campus, Baton Rouge neighborhoods, the Mississippi River watershed, and the Louisiana coast provide authentic local context.
- Students learn science by doing science through a semester-long Course-based Undergraduate Research Experience (CURE).
- ACCESS CloudBank Classroom serves as the primary course science-gateway environment, while SGX3 and other open resources connect students with authentic datasets, maps, computational tools, and research communities.
- Fieldwork and laboratory activities are designed to be low-cost, safe, accessible, and adaptable when weather or transportation interrupts plans.
- Assignments develop workforce-ready skills relevant to Environmental Studies, Geospatial Technologies, and Natural Resources Sustainability.
- Course materials will be open, low-cost, and accessible whenever feasible.

Student Learning Outcomes

1. Explain how ecological, geological, atmospheric, and hydrological processes interact within environmental systems.
2. Analyze relationships among human populations, natural resources, environmental quality, and sustainability.
3. Collect, document, manage, visualize, and interpret environmental field and secondary data.
4. Evaluate environmental hazards using scientific evidence, exposure pathways, dose-response concepts, uncertainty, and risk.
5. Examine how environmental burdens and benefits are distributed across communities.
6. Apply environmental ethics, economics, laws, and policies to authentic environmental decisions.
7. Use geospatial data, CloudBank Classroom, accessible mapping tools, and public environmental databases to investigate environmental patterns.
8. Design and conduct a collaborative environmental research project using responsible and reproducible practices.

9. Communicate environmental findings to scientific, professional, and community audiences.
10. Propose an evidence-based and equitable response to a South Louisiana environmental challenge.

Required OER and Course Resources

Primary OER: Introduction to Environmental Science, 2nd Edition. Available free through the [Open Textbook Library \(opens an external website\)](#). Selected chapters will be assigned rather than requiring students to read the entire book.

Additional open resources may include government datasets and reports, scholarly or community-authored materials, maps, podcasts, videos, and interactive tools from EPA, NOAA, USGS, NASA, Louisiana agencies, iNaturalist, and science gateways. Every assigned source will identify its author or organization, date, purpose, and access requirements.

Students need regular Canvas access, word-processing and presentation software, spreadsheet access, and a device capable of using browser-based notebooks and accessible web maps. The primary computing environment is ACCESS CloudBank Classroom, a managed teaching JupyterHub that provides campus-credential login, individual storage, browser-based tools, and instructor-supported assignment workflows without requiring students to install software or the instructor to maintain a server. Required field equipment will be supplied or replaced with an accessible alternative.

Jetstream2 setup materials will be maintained as an optional reference and contingency resource when additional control, specialized software, or greater computing capacity is needed; Jetstream2 is not the default student platform.

How the Course Works

The course is organized into five modules. Each module follows a predictable learning cycle that moves students from a South Louisiana environmental problem to scientific investigation and action.

Phase	What students do
Engage	Encounter a local case, observation, map, image, dataset, or community question.
Learn	Study environmental concepts, scientific methods, and relevant community perspectives.
Investigate	Collect field evidence or retrieve authentic data through an open science resource.
Analyze	Visualize patterns, evaluate uncertainty, test explanations, and obtain feedback.
Act and reflect	Communicate findings and recommend an equitable, evidence-based response.

Signature Course-Based Undergraduate Research Experience

Jaguar Environmental Resilience CURE

Research focus: How do land cover, vegetation, and the built environment influence urban heat and environmental resilience on Southern University's campus and in surrounding Baton Rouge communities?

The first course pilot will center on urban heat because it supports clear mapping, visualization, field observation, and community interpretation without heavy computation. Research teams may examine relationships among temperature, shade, tree canopy, land cover, impervious surfaces, or neighborhood conditions using gateway data and instructor-approved student observations. Water quality or biodiversity may be used in later semesters after the initial workflow is tested.

The CURE includes student decision-making, discovery, iteration, collaboration, scientific practice, uncertainty, and findings with relevance beyond a single grade. Four scaffolded activities move students through: (1) data discovery, (2) visualization, (3) basic analysis, and (4) interpretation. Formative milestones allow revision before the final report and presentation.

SGX3 and Science Gateway Integration

SGX3 is the NSF-funded Center of Excellence for Science Gateways. In this course, science gateways lower technical barriers by giving students browser-based access to research-quality environmental data, mapping tools, notebooks, and shared workflows. The instructional emphasis is environmental visualization and interpretation rather than advanced programming or computationally intensive analysis.

Role	Student application
Gateway user	Locate, retrieve, document, visualize, or analyze authentic environmental data.
Gateway evaluator	Evaluate data provenance, usability, accessibility, limitations, intended users, and ethical implications.
Gateway designer	Develop a concept for making Louisiana environmental information useful to a defined community.

SGX3 opportunities may also be shared for student training, coding institutes, hackathons, webinars, internships, and the Gateways Conference. Participation in an external opportunity is encouraged but will not be required for course completion unless access is equitable for all students.

Science Gateway Platform and CURE Workflow

- Primary platform: ACCESS CloudBank Classroom, a managed teaching JupyterHub with campus login, persistent student storage, browser-based notebooks, and instructor assignment support.
- Computing on-ramp: A brief HPC Carpentry “Introduction to HPC” experience establishes shared vocabulary about remote and shared computing, files, resources, and responsible use. Students are not expected to run advanced parallel jobs.
- Activity 1 - Discovery: Students locate and document urban-heat, land-cover, elevation, or LiDAR data using the ScienceGateways.org catalog, OpenTopography, and instructor-curated public sources.
- Activity 2 - Visualization: Students adapt an existing tutorial notebook to create an interactive map with leafmap. Geemap may be offered when Google Earth Engine accounts are available and appropriate.
- Activity 3 - Analysis: Students use scaffolded notebooks or spreadsheets to compare patterns, calculate basic statistics, evaluate data quality, and document uncertainty.
- Activity 4 - Interpretation: Students explain environmental patterns, limitations, equity implications, and community relevance. HazMapper may organize geotagged field observations; DesignSafe may provide one optional hazard or resilience dataset without using its engineering-simulation tools.
- Access and continuity: Every notebook-based activity will include a browser-only pathway and a downloadable CSV/spreadsheet fallback. Instructor guides, starter notebooks, sample outputs, and troubleshooting notes will reduce dependence on outside mentors and support reuse in future semesters.

Five Course Modules

Module 1: Reading the Environment

Driving question: How do scientists recognize, measure, and explain patterns in environmental systems?

Timing: Weeks 1–3

- Environmental science, systems thinking, scientific inquiry, and data literacy
- Matter, energy, biogeochemical cycles, ecosystems, biodiversity, and ecosystem services
- Field observations, measurement, metadata, maps, and introduction to science gateways

South Louisiana connection: Southern University campus ecosystems, the Bluff, and the Mississippi River watershed.

Signature work: Campus environmental observation, HPC Carpentry orientation, CloudBank Classroom onboarding, and CURE question bank.

Module 2: People, Place, and Environmental Health

Driving question: Why do environmental hazards and benefits affect communities differently?

Timing: Weeks 4–6

- Human population, consumption, urbanization, land use, and environmental justice
- Environmental toxicology, exposure pathways, dose-response, risk, and uncertainty
- Air, drinking water, waste, contaminated sites, food access, and community health

South Louisiana connection: Baton Rouge air quality, the Mississippi River industrial corridor, neighborhood heat and flooding, and historically underserved communities.

Signature work: Environmental Decision-Making Case and CURE proposal.

Module 3: Water, Wetlands, and the Working Coast

Driving question: How can Louisiana sustain water, wetlands, communities, and livelihoods in a changing environment?

Timing: Weeks 7–9

- Watersheds, hydrology, stormwater, groundwater, and water quality
- Wetlands, ecosystem services, eutrophication, fisheries, and aquatic resources
- Coastal land loss, subsidence, saltwater intrusion, flooding, and restoration

South Louisiana connection: Mississippi River, Baton Rouge drainage, Lake Pontchartrain Basin, wetlands, and coastal communities.

Signature work: Field or open-data investigation, methods approval, and data collection.

Module 4: Energy, Climate, and Resilient Communities

Driving question: How should communities balance energy, employment, health, climate risk, and resilience?

Timing: Weeks 10–12

- Fossil fuels, alternative energy, life-cycle impacts, and Louisiana's energy economy
- Climate science, extreme heat, hurricanes, compound hazards, mitigation, and adaptation
- Urban forests, green infrastructure, risk communication, and equitable resilience

South Louisiana connection: Louisiana energy systems, Baton Rouge heat, hurricanes, and community resilience.

Signature work: Urban-heat visualization and interpretation, Environmental Systems Exam, and preliminary CURE results.

Module 5: Environmental Decisions and Community Action

Driving question: How can environmental professionals turn evidence into responsible and equitable action?

Timing: Weeks 13–16

- Environmental ethics, law, regulation, economics, policy, and stakeholder engagement
- Sustainability, circular economy, urban forestry, conservation, and professional responsibility
- Public communication, reproducibility, accessible data products, environmental careers, and leadership

South Louisiana connection: Industrial permitting, coastal restoration, urban forestry policy, community participation, and the leadership role of HBCU graduates.

Signature work: CURE report, poster, community product, presentation, and Environmental Solutions final.

Assessments and Grading

The course uses a 500-point system. Formative assessments provide structured practice, feedback, revision, and preparation. Summative assessments measure students' independent ability to integrate scientific concepts, environmental data, environmental justice, and professional communication.

Category	Points	Weight
Formative assessments	200	40%
Summative assessments	300	60%
Course total	500	100%

Formative Assessments — 200 Points

Formative work is evaluated for preparation, thoughtful application, documentation, collaboration, and use of feedback. These tasks build toward the major summative products.

Formative assessment	Structure	Points
OER reading checks	8 × 5	40
Environmental field journal	5 × 10	50
CURE data and science-gateway activities	4 × 10	40
CURE research milestones	5 × 10	50
Participation and research-team contributions	Ongoing	20
Formative total		200

CURE Research Milestones

11. Environmental observation and research question
12. Source review and testable hypothesis or research objective
13. Methods, data-management, ethics, accessibility, and safety plan
14. Data collection and research-notebook check
15. Preliminary analysis and research conference

Summative Assessments — 300 Points

Summative assessment	Module	Points
Environmental Systems Exam	4	50
Environmental Decision-Making Case Study	2	50

Summative assessment	Module	Points
CURE Research Report	5	75
CURE Scientific Poster and Presentation	5	50
Community Environmental Product	5	25
Final Environmental Solutions Assessment	5	50
Summative total		300

How the Assessment Categories Work Together

16. Complete a reading check to confirm foundational understanding.
17. Practice observation, field, or data skills in the Environmental Field Journal.
18. Use CloudBank Classroom, an accessible mapping tool, or the CSV/spreadsheet fallback to investigate authentic data and document its source, limitations, and uncertainty.
19. Submit a CURE milestone and use instructor or peer feedback.
20. Complete the related summative product independently or within the approved team structure.

Summative Assignment Descriptions

Environmental Systems Exam — 50 points: One traditional exam assessing foundational concepts, interpretation of data and figures, and application to unfamiliar environmental scenarios.

Environmental Decision-Making Case Study — 50 points: Students analyze a Baton Rouge or South Louisiana environmental decision using scientific evidence, affected populations, uncertainty, stakeholders, trade-offs, and an equitable recommendation.

CURE Research Report — 75 points: Research teams produce a concise scientific report containing the question, context, methods, data, analysis, results, limitations, and significance. Individual accountability evidence is required.

CURE Scientific Poster and Presentation — 50 points: Teams communicate their research through an accessible scientific poster and a brief presentation with questions.

Community Environmental Product — 25 points: Students translate findings into an accessible product for a defined audience, such as a one-page brief, fact sheet, story map concept, infographic, or data-dashboard prototype.

Final Environmental Solutions Assessment — 50 points: Students independently synthesize course concepts to propose and defend a feasible response to a complex environmental scenario. This is an applied final, not a second memorization-heavy exam.

General Evaluation Criteria

- Scientific accuracy and appropriate use of environmental concepts
- Quality, relevance, credibility, and documentation of evidence and data

- Sound methods, reproducible records, and ethical data practices
- Appropriate analysis, visualization, interpretation, and treatment of uncertainty
- Attention to environmental justice, historical context, community knowledge, and stakeholder impacts
- Feasibility, sustainability, and equity of recommendations
- Organization, citation, accessibility, collaboration, and professional communication

Grading Scale

Grade	Points	Percentage
A	450–500	90–100%
B	400–449	80–89%
C	350–399	70–79%
D	300–349	60–69%
F	Below 300	Below 60%

Proposed 16-Week Course Schedule

Week	Module	Focus	Applied work
1	1	Course orientation; environmental science, systems, evidence, safety, and place	Campus case launch; HPC Carpentry introduction; baseline reflection
2	1	Matter, energy, cycles, ecosystems, biodiversity, and ecosystem services	Campus observation and field-journal entry
3	1	Scientific inquiry, environmental measurement, data quality, metadata, and science gateways	CloudBank onboarding; Activity 1 - Discovery; CURE question milestone
4	2	Human population, consumption, urbanization, land use, and environmental justice	Baton Rouge environmental pattern analysis
5	2	Toxicology, exposure pathways, dose-response, risk, and uncertainty	Activity 2 - Visualization; case evidence workshop
6	2	Air, water, waste, food access, cumulative burdens, and community health	Decision-Making Case; CURE hypothesis milestone
7	3	Watersheds, hydrology, groundwater, stormwater, and water quality	Field/data methods workshop
8	3	Wetlands, aquatic ecosystems, eutrophication, fisheries, and ecosystem services	CURE methods and safety milestone
9	3	Coastal land loss, subsidence, saltwater intrusion, flooding, and restoration	CURE data collection; Activity 3 - Analysis
10	4	Energy resources, life-cycle impacts, Louisiana's energy economy, and sustainability	CURE notebook milestone
11	4	Climate science, heat, hurricanes, compound hazards, mitigation, and adaptation	Environmental Systems Exam
12	4	Urban forests, green infrastructure, resilience, and risk communication	Activity 4 - Interpretation; preliminary analysis milestone
13	5	Environmental ethics, law, policy, economics, and stakeholder participation	Results workshop and research conference
14	5	Sustainability, circular economy, conservation, urban forestry, and professional responsibility	Draft report and poster peer review
15	5	Environmental communication, accessibility, careers, and reproducible gateway workflows	CURE Report; Community Product

Week	Module	Focus	Applied work
16	5	Jaguar Environmental Resilience Research Showcase and course synthesis	Poster Presentation; Environmental Solutions Final

Schedule note: Exact readings, due dates, university holidays, examination dates, field locations, and community-partner activities will be added after the term and delivery format are confirmed. Field activities may be rescheduled or replaced when weather, air quality, transportation, accessibility, or safety conditions require it.

Participation and Research Community

Environmental research depends on preparation, reliability, respectful collaboration, and transparent documentation. Students are expected to contribute to team planning, field or data work, analysis, peer feedback, and presentation. Participation may occur through class discussion, Canvas activities, field notes, CloudBank or spreadsheet exercises, research conferences, team records, polls, or reflections so students have multiple ways to contribute.

Attendance and Engagement

Consistent engagement is essential because field investigations, laboratories, team research, and data workshops cannot be fully recreated by reading notes. Attendance and participation will be handled according to university policy. Students should communicate as early as possible about documented absences, caregiving responsibilities, transportation problems, technology limitations, weather emergencies, or other barriers. Reasonable alternatives will focus on demonstrating the missed learning outcome without compromising safety or research integrity.

Late Work, Feedback, and Revision

The course uses staged research assignments so feedback can improve later work. Short module activities should be completed during the assigned module. CURE milestones may be revised as part of the normal research process. Major summative products may be revised once after feedback when submitted by the published revision deadline, except the final assessment. A term-specific late-work window and any deductions will be published before the course begins.

Field, Laboratory, and Research Safety

- Students must complete required safety preparation before participating in field or laboratory work.
- No student should enter private property, industrial sites, hazardous locations, floodwater, moving water, traffic lanes, or restricted areas for a course assignment.
- Students must use assigned protective equipment and follow instructor-approved methods for sampling, transport, storage, and disposal.

- Students should not collect people’s names, health information, photographs, interviews, or other personal data without specific instructor approval and an approved educational protocol.
- Accessible, weather-safe, and non-field alternatives will assess the same learning outcomes when needed.

Ethical Use of Artificial Intelligence

Because AI is a workplace tool that can support environmental data work while also introducing errors, bias, confidentiality concerns, and resource impacts, each assignment will be labeled AI prohibited, AI limited, or AI permitted.

Category	Meaning
AI prohibited	Complete the work without generative AI so independent reading, calculation, observation, reasoning, or reflection can be assessed.
AI limited	AI may support only the listed tasks, such as brainstorming search terms, checking code syntax, or improving clarity; it may not replace source reading, data analysis, or interpretation.
AI permitted	AI may be used as directed, with verification, disclosure, appropriate citation or acknowledgment, and required process evidence.

- Students remain responsible for accuracy, bias, source quality, confidentiality, copyright, data integrity, and the environmental consequences of tool use.
- AI output is not a credible scientific source by itself. Factual claims must be verified using original data or appropriate authoritative sources.
- Students may not fabricate, alter, or silently replace field observations, measurements, images, citations, code outputs, or research records with AI-generated content.
- When AI use is allowed, students must briefly disclose the tool, purpose, prompts or process when requested, and how the output was verified or changed. Every AI-vulnerable summative assignment will identify its AI tier (prohibited, limited, or permitted) and require a short AI-Use Disclosure stating: tool used or “none,” permitted purpose, how output was checked, and what the student changed or rejected. Missing or inaccurate disclosure may be treated as an academic-integrity concern.

Academic Integrity and Source Use

Students must distinguish their work from information, language, media, maps, code, data, and ideas created by others. All borrowed material and datasets must be cited using the course-approved style. Fabricated sources, falsified or selectively omitted data, unauthorized collaboration, undisclosed AI-generated work, and plagiarism will be addressed under Southern University procedures.

Accessibility and Inclusive Course Design

Course materials will be prepared using meaningful headings, descriptive links, readable contrast, alternative text, captions or transcripts, accessible tables, and accessible document formats whenever feasible, following WCAG 2.2-informed practices. Students who need accommodations should contact the appropriate Southern University accessibility office and may speak privately with the instructor about barriers affecting participation. Approved accommodations do not reduce learning outcomes; they provide equitable ways to demonstrate them.

Technology, Weather, and Continuity

South Louisiana weather and infrastructure disruptions may affect travel, field conditions, electricity, internet access, food and water access, and caregiving. When university operations or student access are disrupted, the instructor will communicate through Canvas and university email, adjust deadlines and field plans as appropriate, and prioritize essential learning outcomes. Students should download critical readings, CSV datasets, and assignment instructions when severe weather is anticipated. Notebook activities will include spreadsheet or static-output alternatives when bandwidth, device access, account approval, or platform availability creates a barrier.

Community-Based and Sensitive Topics

Some course topics involve pollution, illness, environmental racism, poverty, food insecurity, disasters, displacement, and communities near students' homes. Students must avoid extracting stories, photographing people, or collecting personal data without permission. Any interviews, community observations, or public-facing activities will include ethical guidance, safety expectations, accessible alternatives, and instructor approval. Course assignments are educational activities and do not constitute human-subjects research.

Data Ethics, Privacy, and Reproducibility

- Preserve original observations and datasets; document cleaning, exclusions, transformations, and corrections.
- Use descriptive filenames, metadata, units, dates, locations at an appropriate precision, and version control appropriate to the assignment.
- Do not publish sensitive species locations, private-property details, personal information, or community data that could create harm.
- Report null results, uncertainty, limitations, and conflicting evidence honestly.
- Follow the license, attribution, and terms of use for every dataset, map, image, gateway, or software tool.

Student Support and University Policies

Before adoption, this section must be replaced or supplemented with the current, officially approved Southern University statements and links for academic integrity, disability/accessibility services, Title IX, nondiscrimination, emergency procedures, counseling, basic-needs support, student conduct, laboratory and field safety, and technology use. Current institutional language should be copied from the official syllabus template rather than recreated from memory.

Instructor Communication

- Announcements: [Canvas and Southern University email]
- Expected response time: [Insert business-day response standard]
- Feedback timeline: [Insert expected turnaround for short and major assignments]
- Office hours and appointments: [Insert location, virtual option, and scheduling process]

Items to Finalize Before Publication

- Instructor, term, meeting information, delivery format, office hours, and required contact information
- Current Southern University required policy statements and student-support links
- Exact readings and confirmation of each resource's open license or lawful access
- Exact due dates, holidays, final-examination requirements, and late-work window
- Approved citation style and templates for the field journal, CURE report, poster, and community product
- CURE topic, available equipment, sampling sites, transportation, accessibility, safety, and data-management plan
- CloudBank Classroom allocation and course dates; approved gateways, datasets, starter notebooks, CSV fallbacks, account requirements, and contingency resources if a platform becomes unavailable
- Any community-partner permissions, public dissemination plan, or research-review requirements

Selected Open and Professional Resources

- [Primary OER textbook - Introduction to Environmental Science, 2nd Edition](#)
- [CUREnet collection of course-based undergraduate research experiences](#)
- [ACCESS CloudBank Classroom access and training](#)
- [Jetstream2 setup tutorial - optional reference and contingency resource](#)
- [HPC Carpentry - Introduction to high-performance computing resources](#)
- [OpenTopography environmental elevation and LiDAR datasets](#)
- [Geemap interactive geospatial mapping notebooks](#)
- [Leafmap lower-barrier interactive mapping notebooks](#)
- [HazMapper collaborative geotagged field-observation mapping](#)
- [DesignSafe natural-hazard and resilience data](#)
- [ScienceGateways.org gateway catalog](#)